Software Requirements Specification for:

Student attendance tracking system

Version V4

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Revision History

Name	Date	Reason for Changes	Version
Hlib Bokatov	03.03.2026	Initial version	V1
Hlib Bokatov	15.03.2026	Added Use case Diagram	V2
Hlib Bokatov	24.03.2026	Added UML Class Diagram	V3
Hlib Bokatov	13.04.2026	Added Activity Diagram	V4

1. Introduction

1.1. Purpose

This document specifies the software requirements for student attendance tracking system. This software is designed for automation attendance tracking for university classes

1.2. Document Conventions

Requirements are labelled as:

- FR-xx – for functional requirements
- NFR-xx – for non-functional requirements

All mandatory requirements use the word “shall”

1.3. Intended Audience and Reading Suggestions

This document is intended for:

- Developers – to implement features
- Testers – to verify system correctness
- University administration – to validate requirements

Recommended reading order:

- Introduction
- Overall Description
- System Features
- External Interface Requirements
- System Features/Models
- Non-functional Requirements

1.4. Product Scope

The Students Attendance Tracking System is a web-based application that allows:

- Teachers to mark attendance digitally
- Students to monitor their attendance percentage
- Automatic eligibility calculation for exams
- Generation of attendance reports

The system replaces manual paper-based attendance registers and improves accuracy and accessibility

2. Overall Description

2.1. Product Perspective

Students Attendance Tracking System is a new web application that shall integrate with the university’s existing student information system. It shall use a secured database to store the attendance records

2.2. Product Features

The system shall provide:

- User authentication (Student, teacher, admin)
- Attendance marked by teachers
- Attendance percentage calculator
- Eligibility determination for exams
- Attendance report export (PDF)

2.3. User Classes and Characteristics

The main user classes of the system are:

- Students - the largest user group. They log in to check their attendance, view their percentage per course, and see whether they are eligible for the final exam.
- Teachers - use the system regularly during classes to mark student attendance, edit records when needed, and export reports. They need a simple and fast interface because attendance is usually marked at the beginning of the class.
- Administrators - manage user accounts, courses, and system configuration. This is the smallest group but has the highest privilege level.

Students and teachers are the most important user classes to satisfy, since they interact with the system daily

2.4. Operating Environment

The system shall operate in the following environment:

- Server side: Linux-based server, running a Node.js runtime and a PostgreSQL database.
- Client side: any modern web browser (Chrome, Firefox, Edge, Safari) on desktop or mobile devices.

2.5. Design and Implementation Constraints

- All communication between the client and server shall be encrypted via HTTPS.
- The system shall be implemented as a web application (no native mobile app in this version).
- PostgreSQL shall be used as the database engine.
- Source code shall follow clean code principles and be maintained in a Git repository.

2.6. Assumptions and Dependencies

- It is assumed that the university provides a stable internet connection and a hosting environment for the system.
- It is assumed that each student and teacher has a valid university account.
- Any change in the university's authentication system may require adjustments on the side of this application.

2.7. UML Class Diagram

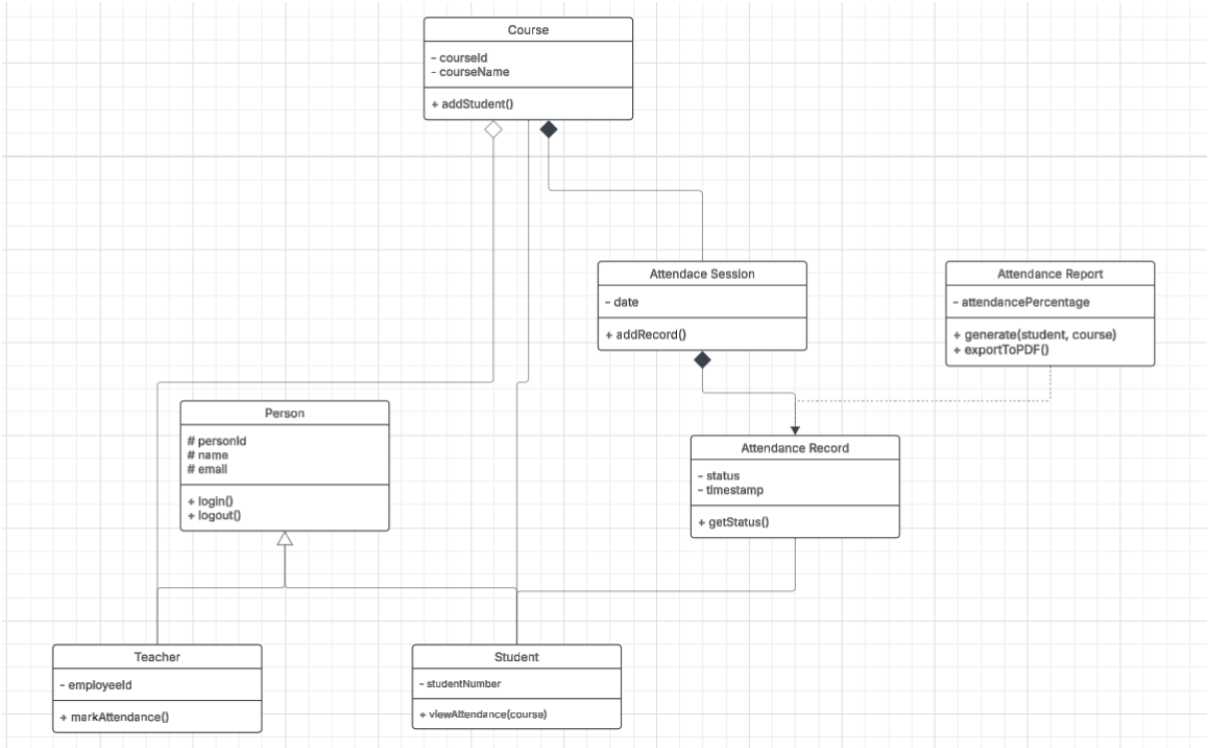

Figure 1. UML Class Diagram

3. System Features

3.1. Use Case Diagram

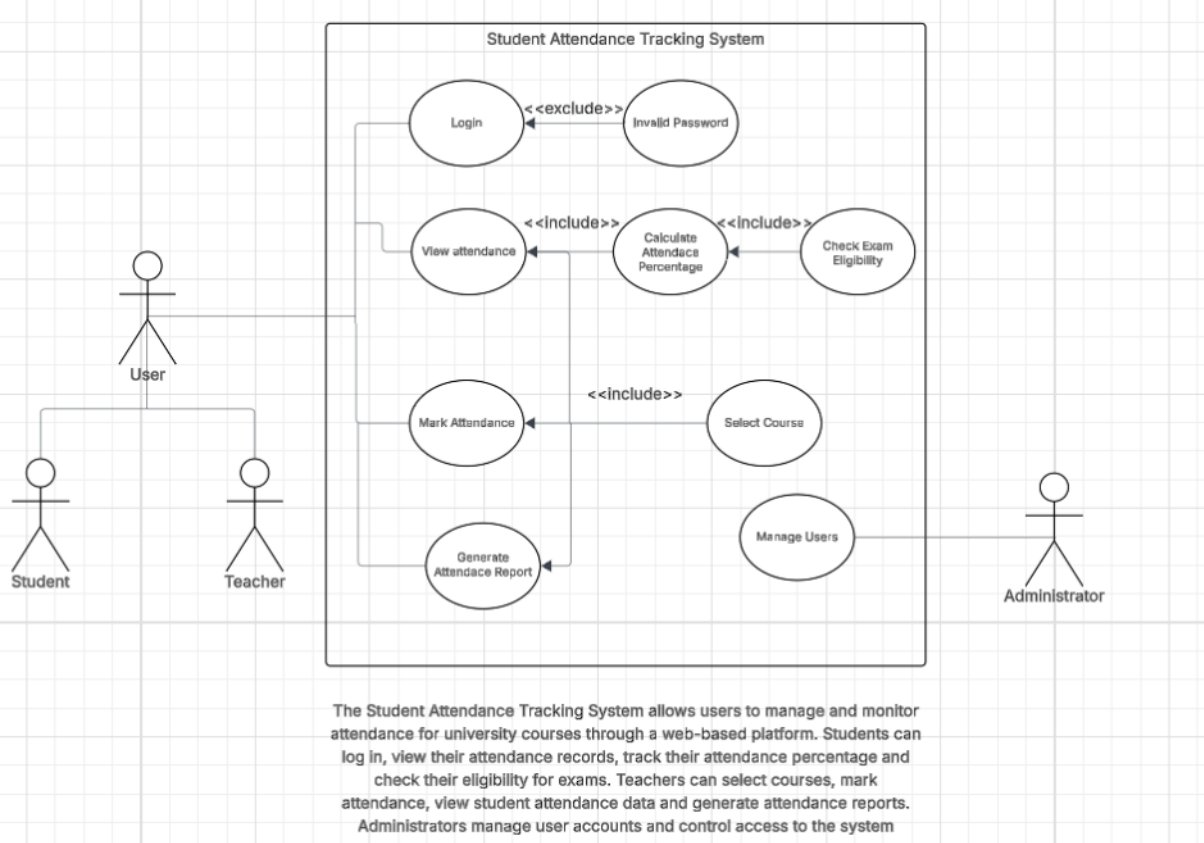

Figure 2. Use Case diagram

3.2. User Authentication

Allows students, teachers, and administrators to log in with their university credentials and access features according to their role.

3.3. Attendance Marking

Allows teachers to mark students as present, absent, or justified during classes and save the results to the database.

3.4. Attendance Overview

Allows students to view their attendance percentage per course and check if they meet the eligibility threshold for the exam.

3.5. Report generation

4. External Interface Requirements

4.1. User Interfaces Overview

4.2. Hardware Interfaces

4.3. Software Interfaces

5. System Features/Modules

5.1.1 Description and Priority

5.1.2 Stimulus/Response Sequences

5.1.3. Activity Diagram

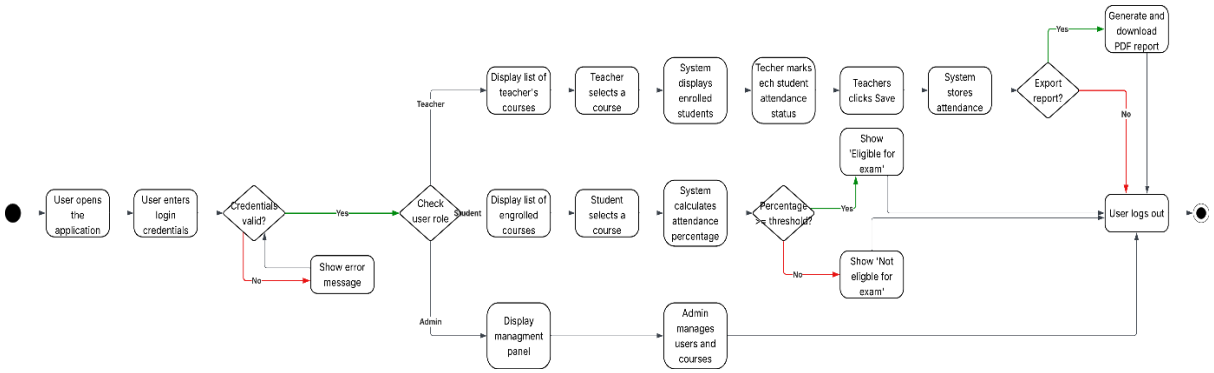

Figure 3. Activity Diagram

5.1.4 Functional Requirements

FR-1.1: When entering valid login credentials the system shall authenticate the user and allow access to the system

FR-1.2: When entering invalid login credentials, the system shall display an error message and deny access

FR-1.3: The system shall differentiate user roles (student or teacher) and display the appropriate content

FR-1.4: The system shall allow users to log out of the system securely

FR-1.5: When entering valid login credentials, the system shall authenticate the user and allow access to the system

FR-1.6: When selecting a course, the system shall display the list of students enrolled in that course

FR-1.7: The system shall store attendance records in the database after submission

FR-1.7: The system shall allow the teacher to mark students as present, absent or justified

6. Nonfunctional Requirements

6.1. Performance

NFR-1.1: The system shall respond to user requests withing 2 seconds under normal operating conditions

NFR-1.2: The system shall support at least 200 simultaneous users without performance drop

6.2. Security

NFR-2.1: The system shall require user authentication before accessing private data

NFR-2.2: The system shall restrict access to modifying attendance to authorized teachers only

7. GitHub repository